YOUNG'S DOUBLE SLIT EXPERIMENT

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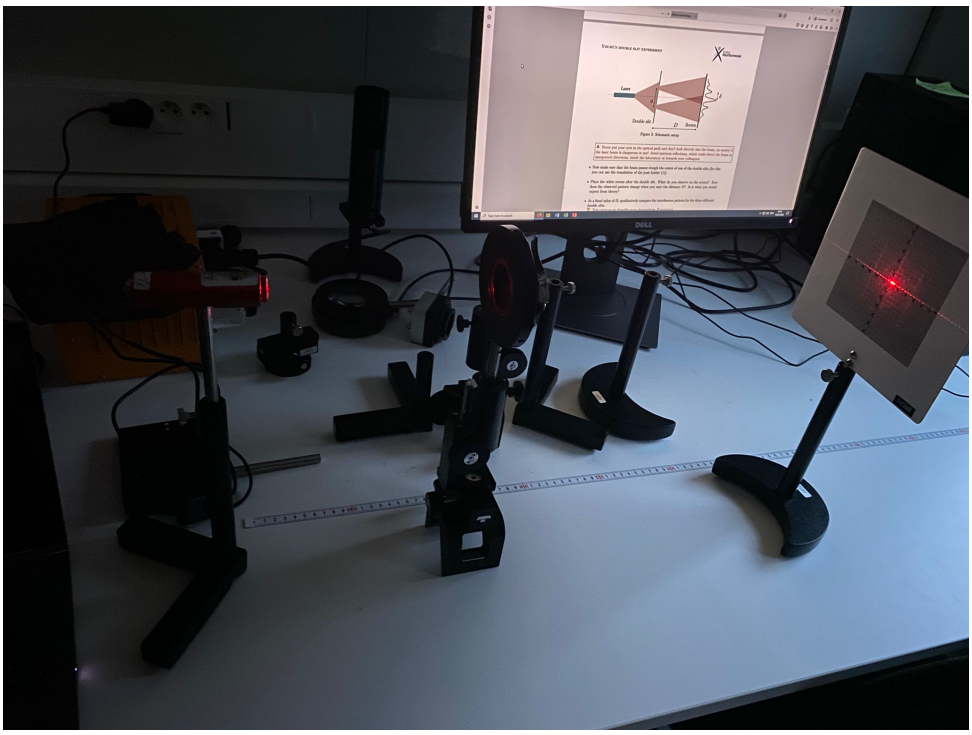

Figure 1: Optical setup: source (red laser), double slit disk, screen

The goal of this lab session is to set up the double slit experiment and observe light interferences, then to exploit this phenomenon in order to measure the wavelength of different light sources.

1 INTERFERENCE FRINGES AND WAVELENGTH MEASUREMENT

In this part, we reproduced Young's experiment and observed interference fringe from which we measured the wavelength of the source. Our sources for this part are: a red laser, a blue laser, and a sodium spectral lamp

1.1 Preliminary argument and observations

We made sure that our source and screen were correctly aligned, by progressively separating them while keeping the alignment, and the placed the double slit element in between our laser and white screen, obtaining the following diffraction pattern. As we can see in the image, we

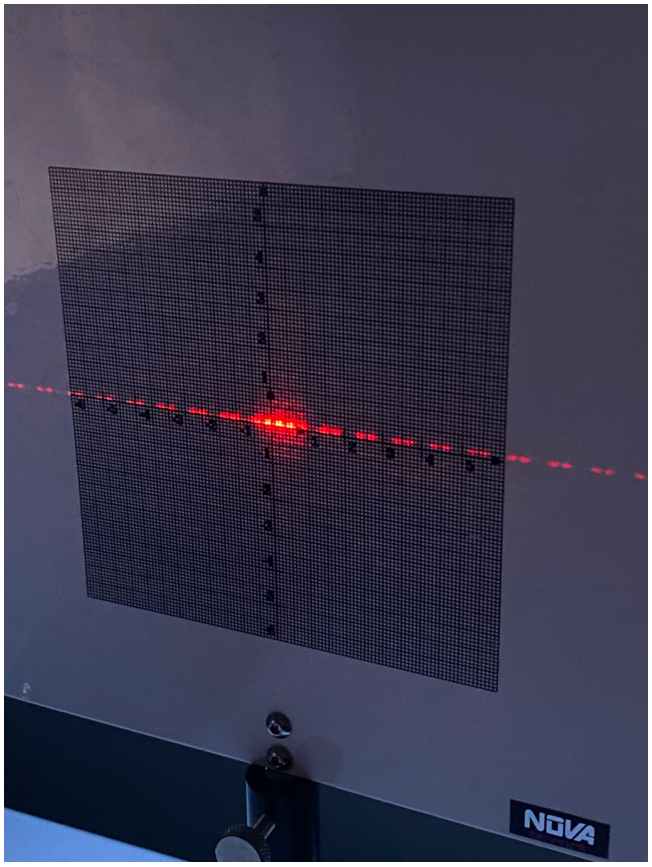

Figure 2: Obtained double slit diffraction pattern

obtain a regular pattern diffraction pattern, however there are also thin vertical interference fringes.
Our observed pattern changes in the following manner when we vary D, the distance between the screen and the slits: as D increased, so did the fringe spacing; when slit separation increased for a fixed D, the fringe spacing decreased. We are given:

$$\delta = \frac{\lambda D}{a}$$

Hence the fringe spacing δ is proportional to D and inversely proportional to a, the spacing between the slits.

1.2 Quantitative exploitation

For this part we aim to assess experimentally the validity of the formula $\delta = \frac{\lambda D}{a}$. Although, in theory, the aperiodicity of the interference figure could be adjusted by using narrower slits, it would not be practical: thinner slits transmit much less light, making the interference pattern too dim to measure accurately. In addition, narrow slits are more difficult to align properly.
In the following sections, we measure the interfringe distance while varying the parameters D and a, and we reproduce the measurements for both lasers and a sodium spectral lamp.

1.2.1 • Red and Blue lasers

All measurements in this section were done with a CMOS detector (better to avoid the blooming effect which distorts the signal shape) equipped with a neutral filter to avoid saturation which would corrupt our measurements. For each laser, we always measured δ for three different slits spacing a at three different distances D. Below is all our data:

$\frac{D}{a}$ (dimensionless)	δ_{red} (μm)	δ_{blue} (μm)
1800	1121	736.8
3000	1892	1201.333333
4500	2768	1814
3500	2102.666667	1456
2333.333333	1388	918
1400	838	576
1000	602.8571429	390
1666.666667	1001.6	752
2500	1500	918.6666667

Table 1: Experimental measurements of fringe spacing δ as a function of D/a.

Sources of errors. The dominant uncertainty in this experiment comes from the measurement of the distance D between the double slit and the CMOS camera.D was measured using a long meter on the table so the measurement was effectively taken along the table rather than along the true optical axis, which introduces a significant geometric error. A second important source of uncertainty is the possible angular misalignment between the double slit and the camera (the camera not being perfectly perpendicular to the propagation direction), which can slightly distort the observed fringe spacing (we tried minimizing this error by looking at the reflection of the laser on the camera and centering it). Finally, the fringe spacing δ has an experimental uncertainty due to the finite spatial sampling of the CMOS pixels and the manual selection of minima on the softare. This was reduced by measuring δ over N fringes and dividing by N.

Uncertainty on $X = D/a$. The slit separation a is provided with an uncertainty $u_a \approx 2\ \mu$m. The distance D between the double slit and the detector is measured using a meter. Since the measurement is not performed on an optical rail and the planes of the slit plate and detector are not perfectly defined, we take a conservative uncertainty $u_D \approx 2$ cm $= 2 \times 10^4\ \mu$m. For the dimensionless quantity $X = D/a$, uncertainty propagation gives

$$u(X) = X\sqrt{\left(\frac{u_D}{D}\right)^2 + \left(\frac{u_a}{a}\right)^2} = \frac{D}{a}\sqrt{\left(\frac{2\times 10^4}{D}\right)^2 + \left(\frac{2}{a}\right)^2},$$

where D and a are expressed in μm.

Uncertainty on the fringe spacing δ. The interference pattern is recorded by a CMOS sensor with 2048 pixels over a length $L = 8.2$ mm, giving a spatial sampling step along the measurement axis

$$p = \frac{L}{2048} \approx 4.0\ \mu\text{m}.$$

The fringe spacing is extracted by manually selecting two minima separated by N fringes and using $\delta = (x_{m+N} - x_m)/N$. During our experiment, the minima were not perfectly sharp: detector noise and the temporal/spatial smoothing broaden the dark fringes, making the exact location of the minimum not very precise over several pixels. To account for this, we adopt a conservative manual selection uncertainty of 50 pixels,

$$u_{\text{click}}(x) \approx 50\,p \approx 200\ \mu\text{m}.$$

The uncertainty on the pixel resolution of the camera can be neglected so the total uncertainty on the position of a minimum is

$$u(x) \approx u_{click} \approx 200\mu\text{m}.$$

Finally, uncertainty propagation on $\delta = (x_{m+N} - x_m)/N$ yields

$$u(\delta) = \frac{\sqrt{2}\,u(x)}{N} \approx \frac{282}{N}\ \mu\text{m}.$$

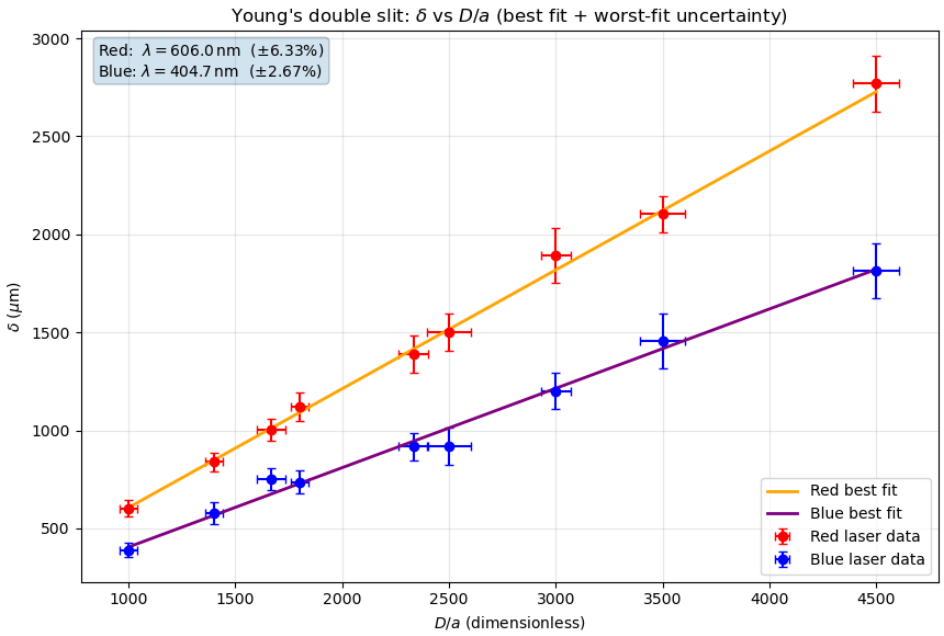

Figure 3: $\delta(\mu m)$ vs $\frac{D}{a}$ with linear fit for both lasers

Finally, the estimated wavelength λ is obtained by fitting the values of δ against D/a. To get an uncertainty on lambda, we used a best fit/ worst fit method which we coded in python.

Red laser :

$$\lambda_{\text{red}} = (606 \pm 6.33\%)\, nm = (606 \pm 39)\, nm.$$

Blue laser :

$$\bar{\lambda}_{\text{blue}} = (404.7 \pm 2.67\%)\, nm = (404.7 \pm 11)\, nm.$$

The fitted wavelengths are $\lambda_{\text{red}} = (606 \pm 39)\,\text{nm}$ and $\lambda_{\text{blue}} = (404.7 \pm 11)\,\text{nm}$. Comparing with the expected values $\lambda_{\text{red,th}} = 632\,\text{nm}$ and $\lambda_{\text{blue,th}} = 405\,\text{nm}$, the relative deviations are

$$\frac{|\lambda_{\text{red}} - \lambda_{\text{red,th}}|}{\lambda_{\text{red,th}}} = \frac{|606 - 632|}{632} \approx 4.1\%, \qquad \frac{|\lambda_{\text{blue}} - \lambda_{\text{blue,th}}|}{\lambda_{\text{blue,th}}} = \frac{|404.7 - 405|}{405} \approx 0.07\%.$$

The blue laser result is therefore in excellent agreement with the theoretical wavelength, well within the experimental uncertainty. For the red laser, the measured value is lower than expected and the discrepancy ($\sim 4.1\%$) is comparable to the uncertainty estimate ($\sim 6.3\%$), so the result remains compatible with the theoretical value within error bars.

1.2.2 • Sodium spectral lamp

For this experiment we used a sodium spectral lamp. Here the condenser lens is necessary. Indeed, unlike the laser our light is not collimated, but rather its coming from a relatively large emitting area of independent and incoherent sources, and they would each produce their own interference pattern. So the condenser lens essentially creates an effective point source which is

needed for the experiment. The width of our first single slit has to be large enough to allow for a sufficient quantity of light to pass through but thin enough for diffraction to happen. To take our measurements, we switch to the CCD detector, which is more sensitive than the CMOS detector, because the intensity of the light emitted by the spectral lamp is very low.

Sources of errors. This part of the experiment was significantly more delicate than with the laser diodes, mainly because the sodium lamp is an extended and weak source. As a result, the fringe contrast was low and the signal was strongly affected by ambient light and detector noise, which made the localisation of minima difficult. In addition, the quality of the interference pattern is very sensitive to alignment: the slit (S) must be strictly parallel to the double slit (DS), and any small angular misalignment reduces the spatial coherence and rapidly washes out the fringes. The condenser lens also introduces possible systematic errors (imperfect focusing, aberrations and slight misplacement), which can modify the effective source size seen by the double slit and therefore the visibility of the pattern. Finally, the distance d between the slit (S) and the double slit (DS) must be sufficiently large to ensure that the illumination of the double slit is approximately uniform, if d is too small, the double slit sees a larger angular extent of the source, reducing coherence and fringe visibility. Because of these limitations, the fringe spacing δ was often extracted over only one period, which directly affected the precision compared to the laser measurements.

Uncertainties. The wavelength is obtained from the same relation $\delta = \lambda D/a$, hence $\lambda = \delta\, a/D$. We keep the same uncertainties as in the laser section for the geometric quantities, namely $u_a \approx 2\ \mu$m and $u_D \approx 2$ cm (measurement performed with a meter, without a rigid rail). The dominant change in this section comes from the uncertainty on δ: due to the low contrast and strong noise, the minima were poorly defined, and we were generally unable to average over several fringes.

$$u(\delta) = \frac{\sqrt{2}\, u(x)}{N},$$

where $u(x)$ accounts for the detector sampling and the manual selection of minima. Consequently, the final uncertainty on λ is significantly larger than for the laser sources and is mainly limited by the precision on δ rather than by the geometric measurements.

Ultimately, we were only able to pick 3 acceptable measurements to plot, as most of our measurements were simply taking the minima cause by the distance between the slits into account, and we not taking actual interference fringes.

Result :

$$\lambda = 547.6 \pm 322nm$$

We obtained $\lambda = (548 \pm 322)$ nm. The expected wavelength of a sodium lamp is $\lambda_{\mathrm{Na}} \approx 589$ nm, which lies within our uncertainty interval, so our result is compatible with the theoretical value. However, the relative uncertainty is very large ($322/548 \simeq 59\%$), mainly due to the

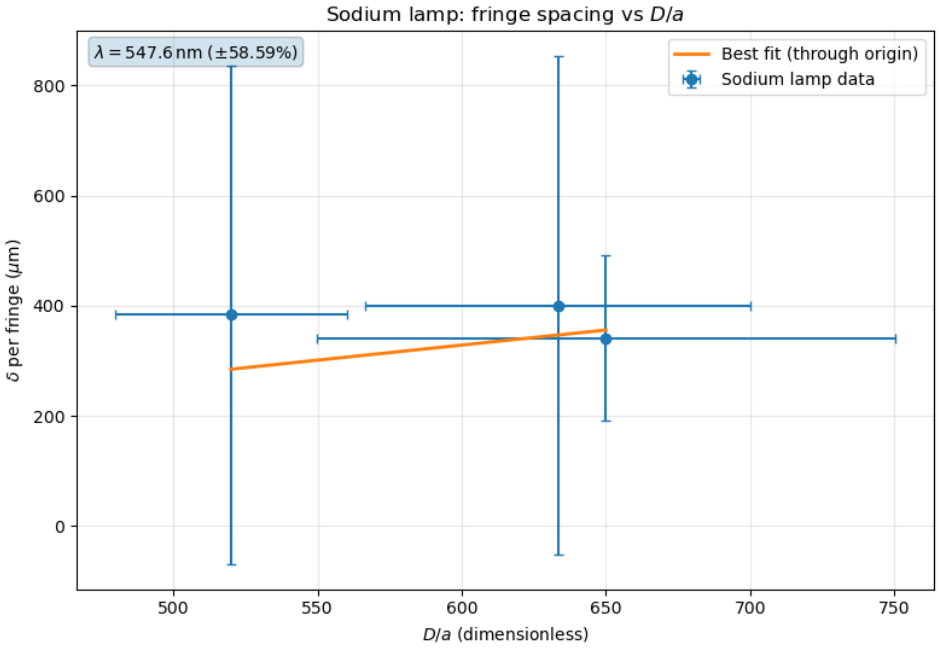

Figure 4: Plot of δ for the spectral lamp

low fringe contrast and noise, and because δ was often extracted over only one period ($N \simeq 1$), which strongly degrades the precision.

Figure 5: Optical setup: sodium lamp, condenser lens, single slit, double slit, CCD camera

2 SPATIAL COHERENCE AND LOSS OF CONTRAST

Theory gives us that the fringe contrast is proportional to

$$C = \mathrm{sinc}\left(\frac{\pi a b}{\lambda d}\right),$$

and the first zero contrast occurs when:

$$b = \frac{\lambda d}{a}. \tag{1}$$

This provides an independent method to estimate λ using the disappearance of fringes.

2.1 Qualitative experiment

- Moving the slit in the direction perpendicular to the incoming light beam, we observe that our fringes moved in the opposite direction. This can be explained by the fact that

when the slit if shifted to one side, rays going to the the opposite side of the screen travel a shorter distance, and rays going to the same side travel a longer distance.

- With the graduation that was on the stand, we found that a displacement of S of 1.7 mm moved the fringes by approximately one period.
- By not moving the slit and opening it larger, we see that the contrast of our interference pattern slowly decreases until our fringes disappear totally.
- The slit (S) must be strictly parallel to the double slit (DS) to avoid averaging interference patterns that are shifted differently along the vertical direction. If (S) is tilted, different vertical positions correspond to different effective source points X and therefore to different phase offsets nd then the superposition reduces the visibility and can lead to fringes that are not well resolved by the linear sensor.

2.2 Spatial coherence

The theory tells us predicts that the fringe contrast becomes exactly zero when the slit opening reaches the value

$$b_0 = \frac{\lambda d}{a}, \tag{2}$$

where b_0 is the first zero-contrast width, d is the distance between the single slit (S) and the double slit (DS), a is the separation between the two slits of the double slit, and λ is the wavelength.

For several choices of (d, a),the method was to increase b until the fringes disappear (first zero contrast point). Using Eq. (1), each measurement gives

$$\lambda_i = \frac{b_i\, a_i}{d_i}. \tag{3}$$

For this part, we could expect to obtain a value compatible with our above result obtained from the interfringe measurement (within uncertainty). However, we can also expect the experimental uncertainty to be more important here, since the determination of the zero-contrast point relies on a visual criterion, which is inherently subjective.

Despite time constraints during the lab, we were able to perform a limited quantitative measurement of the zero-contrast condition for several values of the slit separation a, while keeping the distance d fixed. The zero-contrast point was determined visually by increasing the single-slit opening b until the interference fringes disappeared.

The uncertainty on b is dominated by the finite precision of the slit adjustment, while the uncertainties on a and d are comparatively smaller. As a result, the spatial coherence method is less precise than the interfringe measurement, but it still illustrates well the role of spatial filtering and coherence in the formation of interference fringes.

Error and uncertainty analysis

a (μm)	d (cm)	b (mm)	λ (nm)	$\Delta\lambda$ (nm)
500	8.5	0.10	588	325
300	8.5	0.17	600	226
200	8.5	0.27	635	190

Table 2: Zero-contrast measurements and corresponding wavelength values obtained using Eq. (3).

We consider the uncertainties associated with a, d, and b. The uncertainty on the slit opening was taken to be half a graduation on the micrometer screw,

$$\Delta b = 0.05 \text{ mm} = 5 \times 10^{-5} \text{ m}.$$

The uncertainty on a was given in the appendix as

$$\Delta a = 2 \ \mu\text{m}.$$

We determined the uncertainty on d to be

$$\Delta d = 2 \text{ cm}.$$

The uncertainty on the quantity $\frac{b\,a}{d}$ is computed using standard propagation of uncertainties:

$$\Delta\left(\frac{b\,a}{d}\right) = \sqrt{\left(\frac{\partial f}{\partial a}\right)^2 (\Delta a)^2 + \left(\frac{\partial f}{\partial d}\right)^2 (\Delta d)^2 + \left(\frac{\partial f}{\partial b}\right)^2 (\Delta b)^2}, \qquad f = \frac{b\,a}{d}.$$

The mean wavelength is computed as the arithmetic average

$$\bar{\lambda} = \frac{1}{N}\sum_{i=1}^{N} \lambda_i,$$

yielding

$$\bar{\lambda} \approx 608 \text{ nm}.$$

This value is consistent, within experimental uncertainty, with the expected wavelength of the sodium lamp. The relatively large uncertainties reflect the strong sensitivity of the method to the determination of the zero-contrast point, as well as the limited number of data points. Overall, this experiment primarily serves to illustrate the role of spatial coherence in interference phenomena rather than to provide a precise wavelength measurement.

2.2.1 • Bonus : estimating Betelgeuse and the Sun's diameter

The disappearance of fringes for an inter-slit distance $a \approx 3\,\text{m}$ indicates that the angular diameter of the source is of the order of the coherence angle:

$$\theta_\star \sim \frac{\lambda}{a}.$$

Assuming Betelgeuse is at distance $d_\star$, its physical diameter is

$$D_\star \sim \theta_\star \, d_\star \sim \frac{\lambda}{a} \, d_\star.$$

We use:

$$a = 3\,\mathrm{m}, \qquad \lambda = 6.0 \times 10^{-7}\,\mathrm{m}, \qquad d_\star \approx 600\,\mathrm{ly}.$$

Convert light-years to meters (with $1\,\mathrm{ly} \approx 9.46 \times 10^{15}\,\mathrm{m}$):

$$d_\star \approx 600 \times 9.46 \times 10^{15} \approx 5.68 \times 10^{18}\,\mathrm{m}.$$

Then

$$D_\star \sim \frac{6.0 \times 10^{-7}}{3}\,(5.68 \times 10^{18}) = (2.0 \times 10^{-7})(5.68 \times 10^{18}) \approx 1.14 \times 10^{12}\,\mathrm{m}.$$

for the sun

Using the small-angle relation $D_\odot \approx \theta_\odot \, d_\odot$, with

$$\theta_\odot \approx 9.3 \times 10^{-3}\,\mathrm{rad}, \qquad d_\odot \approx 1.496 \times 10^{11}\,\mathrm{m},$$

we obtain

$$D_\odot \approx (9.3 \times 10^{-3})(1.496 \times 10^{11}) = 1.391 \times 10^{9}\,\mathrm{m} \approx 1.39 \times 10^{9}\,\mathrm{m}.$$

Comparison

The ratio of diameters is therefore

$$\frac{D_\star}{D_\odot} \approx \frac{1.14 \times 10^{12}}{1.39 \times 10^{9}} = \frac{1.14}{1.39} \times 10^{3} \approx 0.82 \times 10^{3} \approx 8.2 \times 10^{2}.$$

So Betelgeuse is of the order of 10^3 times larger in diameter than the Sun.

2.3 Coherent light source

For this last part, our observations were that when the light source was the laser, ajdustment of the (S) slit did not cause major changes, which makes sense because a laser diode is a spatially coherent source, "which means that different parts of the beam produce waves with a well defined phase relationship between them.", contrarily to incoherent sources, notably the sodium lamp we used in the above experiment. So the interference figure is not modified by the slit, because superposition is not from intensity but from amplitude so this keeps the visibility and coherence of our pattern.

3 CONCLUSION

In this lab, we reproduced Young's double slit experiment and verified the linear relation $\delta = \lambda D/a$ by measuring the fringe spacing for different slit separations and screen distances. Using a CMOS detector and a best-fit/worst-fit method, we obtained wavelength estimates $\lambda_{\text{red}} = (606 \pm 39)\,\text{nm}$ and $\lambda_{\text{blue}} = (404.7 \pm 11)\,\text{nm}$, in good agreement with the expected values. We then repeated the measurement with a sodium spectral lamp, highlighting the importance of spatial filtering and improved sensitivity (CCD detector) for extended incoherent sources; although our value was compatible with $\lambda_{\text{Na}} \approx 589\,\text{nm}$, the uncertainty remained large due to the low fringe contrast (and our inability to produce fringes consistently) and measurement noise. Finally, the spatial coherence study illustrated how increasing the source size reduces fringe visibility, and how the disappearance of contrast provides an independent (but less precise) method to estimate the wavelength.